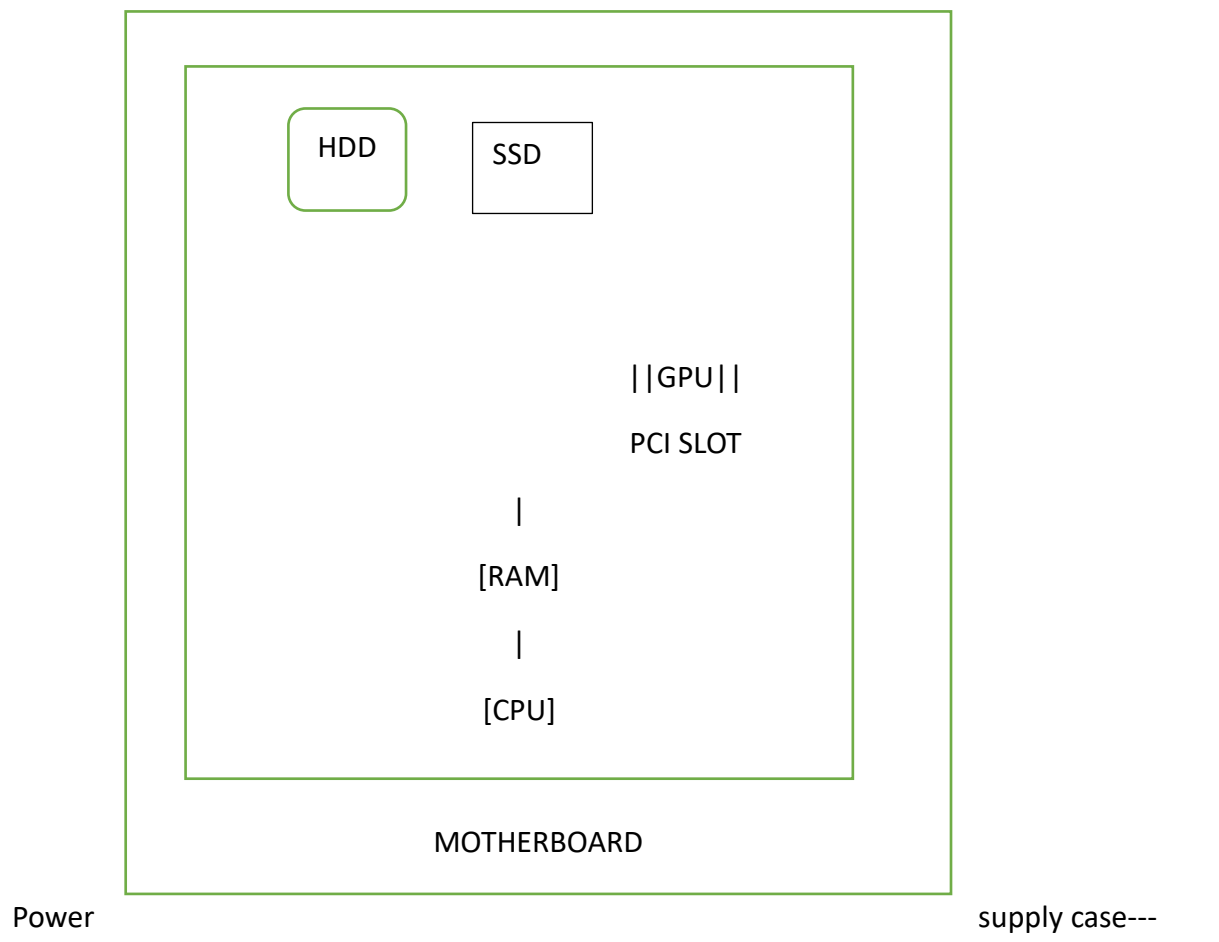


DAY 1 ASSIGNMENT

1. Draw a labeled diagram of a desktop PC, clearly marking the motherboard, CPU, RAM, GPU, and both HDD and SSD storage components.

The labelled diagram is as follow:



DAY 1 ASSIGNMENT

2. Create a comparison table showing the main differences between HDD and SSD storage in terms of speed, durability, and typical use cases.

Hint: Think about which one is faster, which is more shock-resistant, and where you would usually find each type (laptop, gaming PC, etc.).

Feature	HDD (Hard Disk Drive)	SSD (Solid State Drive)
Storage Technology	Spinning magnetic platters + mechanical arm	NAND flash memory (no moving parts)
Speed	80–150 MB/s (slower boot & load times)	500–3,500+ MB/s (instant boot & fast transfers)
Durability	Prone to damage due to moving parts	Highly durable, shock-resistant
Noise	Audible spinning and clicking sounds	Silent operation
Power Consumption	Higher, generates more heat	Lower, energy-efficient
Capacity	Larger (up to 20TB or more) at lower cost	Smaller (commonly 256GB–8TB), higher cost per GB
Cost	Cheaper per GB	More expensive per GB
Best Use Cases	Bulk storage, backups, budget PCs	Gaming, laptops, fast boot systems, professional workstations

3. Write a short explanation (2-3 sentences each) describing the role of the CPU, RAM, and GPU in running a game like PUBG or Free Fire on a PC.

CPU (Central Processing Unit): Acts as the “brain” of the PC. In PUBG or Free Fire, it handles game logic, physics calculations, and AI behavior. A strong CPU ensures smooth frame pacing and prevents bottlenecks

DAY 1 ASSIGNMENT

RAM (Random Access Memory): Provides fast, temporary storage for game data. It keeps textures, maps, and active processes ready for quick access. More RAM reduces lag and allows smoother multitasking while gaming.

GPU (Graphics Processing Unit): Renders visuals, effects, and animations. In games like PUBG, the GPU determines frame rates, resolution, and graphical quality. A powerful GPU is essential for realistic graphics and smooth gameplay.

4. Imagine you want to upgrade your PC to run the latest version of FIFA smoothly. List which hardware components you would consider upgrading and explain why for each.

-GPU (Graphics Card): Upgrade to a modern mid/high-end GPU (e.g., **NVIDIA RTX 3060** or **AMD RX 6600**). FIFA relies heavily on graphics rendering for stadiums, players, and crowd details.

-CPU: A quad-core or higher processor (e.g., Intel i5/i7 or AMD Ryzen 5/7) ensures smooth physics, AI, and match simulations.

- RAM: At least **16 GB DDR4/DDR5** is recommended for modern games to avoid stuttering and allow background tasks.

-Storage: Install FIFA on an **SSD** for faster load times, quick boot, and reduced lag compared to HDD.

-Cooling & Power Supply: Ensure adequate cooling and a reliable PSU to support upgraded components and prevent overheating.
